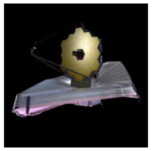

# 4570 - Moving beyond the Milky Way: Enabling Cross-Observatory Proper Motion Determinations with HST and JWST

Cycle: 2, Proposal Category: GO

**INVESTIGATORS**


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**OBSERVATIONS**

| Folder | Observation | Label | Observing Template | Science Target |
|---|---|---|---|---|
| Observation Folder | | | | |
| | 1 | Draco II | NIRCam Imaging | (1) NAME-DRACO-II |
| | 2 | WLM | NIRCam Imaging | (2) WLM |

**ABSTRACT**

HST has made unique contributions to our understanding of the dynamical evolution of the Milky Way and its constituent satellites, globular clusters, halo stars and tidal streams. Proper motions enabled by multi-epoch HST observations have been used to probe the nature of dark matter, the epoch of reionization, and correlated satellite infall. While Gaia now, too, provides unparalleled astrometric precision for the Milky Way, at outer halo distances this is only for intrinsically bright stars. Moving beyond the Milky Way halo will only be possible by utilizing long baselines with HST or by extending these baselines using another facility, such as JWST or Roman, in tandem with a first epoch from HST. Thus, it is imperative that we calibrate cross-observatory techniques to measure absolute proper motions. Here, we request 8 orbits with ACS/WFC and 3.27 hours with JWST's NIRCam for Draco II, an ultra-faint dwarf galaxy, and WLM, a star-forming galaxy. Both objects already have an epoch of ACS/WFC data, as well as publicly available NIRCam data from the Resolved Stellar Populations Early Release Science program. We will use the newly requested data to measure an HST-HST proper motion for each dwarf. This proper motion will be used as a "ground truth" to investigate and calibrate (i) detector-based systematics and (ii) reference frame-based systematics, in combining HST and JWST data. We stress the urgency of getting these observations now while HST is operating well. Such cross-observatory calibrations will be extremely difficult if the astrometric capabilities of HST degrade.

**OBSERVING DESCRIPTION**

We will observe Draco II - an ultra-faint dwarf galaxy - and WLM - a star-forming, isolated dwarf galaxy - with NIRCam. Both objects already have an epoch of HST/ACS data, as well as publicly available NIRCam data from the Resolved Stellar Populations Early Release Science program. One method of measuring proper motions is to build a stationary reference frame using background galaxies. In the case of a cross-observatory measurement between HST and JWST, this process is prone to systematic errors unless the evolution of galaxy profiles across instruments is fully understood. We will observe these fields deeply, with a SNR of about 23 at F090W = 27, using the F200W filter, which is the NIRCam filter for which the PSF is best sampled. The observed fields around these galaxies have been chosen to exactly overlap the fields NIRCam previously observed in F090W and F150W, allowing us to track the evolution of galaxy profiles with wavelength. These observations coincide with second-epoch HST/ACS obvervations, which will be used in an HST-HST PM measurement for calibrating our HST-JWST PM measurement.

Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | NAME-DRACO-II | RA: 15 52 47.5920 (238.1983000d)<br>Dec: +64 33 55.08 (64.56530d)<br>Equinox: J2000 | | |
| *Comments:*<br>*Category=Galaxy*<br>*Description=[Dwarf galaxies]* | | | | |
| (2) | WLM | RA: 00 01 57.3000 (.4887500d)<br>Dec: -15 31 22.70 (-15.52297d)<br>Equinox: J2000 | | |
| *Comments:*<br>*Category=Galaxy*<br>*Description=[Dwarf irregular galaxies]* | | | | |

## Observation

**Proposal 4570, Observation 1: Draco II**

Thu Sep 14 22:01:04 GMT 2023

**Diagnostic Status: Warning**

Observing Template: NIRCam Imaging

## Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | NAME-DRACO-II | RA: 15 52 47.5920 (238.1983000d)<br>Dec: +64 33 55.08 (64.56530d)<br>Equinox: J2000 | | |

*Comments:*
*Category=Galaxy*
*Description=[Dwarf galaxies]*

## Template

| Module | Subarray | Target Placement |
|---|---|---|
| ALL | FULL | Module Gap |

## Dithers

| # | Primary Dither Type | Primary Dithers | Subpixel Dither Type | Dither Size | Subpixel Positions |
|---|---|---|---|---|---|
| 1 | NONE | | STANDARD | | 4 |

## Spectral Elements

| # | Short Filter | Long Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Dithers | Total Exposure Time | ETC Wkbk.Calc ID |
|---|---|---|---|---|---|---|---|---|---|
| 1 | F200W | F277W | MEDIUM8 | 7 | 2 | 8 | 4 | 5883.75 | |

## Special Requirements

Aperture PA Range 142.4811776496318 to 142.4811776496318 Degrees (V3 142.5525307496318 to 142.5525307496318)

## Observation

**Proposal 4570, Observation 2: WLM**

Thu Sep 14 22:01:04 GMT 2023

**Diagnostic Status: Warning**

Observing Template: NIRCam Imaging

## Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 2:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (2) | WLM | RA: 00 01 57.3000 (.4887500d)<br>Dec: -15 31 22.70 (-15.52297d)<br>Equinox: J2000 | | |

*Comments:*
*Category=Galaxy*
*Description=[Dwarf irregular galaxies]*

## Template

| Module | Subarray | Target Placement |
|---|---|---|
| ALL | FULL | Module Gap |

## Dithers

| # | Primary Dither Type | Primary Dithers | Subpixel Dither Type | Dither Size | Subpixel Positions |
|---|---|---|---|---|---|
| 1 | NONE | | STANDARD | | 4 |

## Spectral Elements

| # | Short Filter | Long Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Dithers | Total Exposure Time | ETC Wkbk.Calc ID |
|---|---|---|---|---|---|---|---|---|---|
| 1 | F200W | F360M | MEDIUM8 | 7 | 2 | 8 | 4 | 5883.75 | |

## Special Requirements

Aperture PA Range 259.5499223275193 to 259.5499223275193 Degrees (V3 259.6212754275193 to 259.6212754275193)